

A square-root obstruction

to evaluating increasing

Hahn series at ω

A negative answer to Lipparini's Problem 7.7,
with a nonnegative-coefficient semiring strengthening,
infinitesimal evaluation and monomial classification

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The decisive identity already lies in $\mathbb{Q}[[x]]$:

$$B(x) = 1 - \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{16} - \frac{5x^4}{128} - \cdots, \quad B(x)^2 = 1 - x.$$

A ring homomorphism sending x to ω would give

$$\Phi(B)^2 = 1 - \omega < 0,$$

which is impossible in the ordered field of surreal numbers. Subtraction is not to blame: the all-positive pair $H(x) = \sum_{n \geq 0} \binom{2n}{n} 4^{-n} x^n$ and $G(x) = \sum_{n \geq 0} x^n$ satisfies $H^2 = G$ and $G = 1 + xG$, so even a homomorphism of the *semiring* $\mathbb{Q}_{\geq 0}[[x]]$ cannot send x to any $u \geq 1$.

Abstract. We give a negative answer to the increasing-support Hahn-series evaluation problem posed as Problem 7.7 in the April 2026 extended version of Paolo Lipparini's paper on monotone infinitary ordinal operations. The obstruction is elementary and algebraic: the formal square root of $1 - x$ exists in the source, whereas $1 - \omega$ has no square root in an ordered field. No continuity, order preservation, or preservation of infinite sums is assumed. We then strengthen the obstruction along two axes at once: it survives on the semiring $\mathbb{Q}_{\geq 0}[[x]]$ of nonnegative-coefficient series, where there are no additive inverses at all, and there it already excludes every value $u \geq 1$. We prove that the possible images of x under unital homomorphisms $\mathbb{Q}[[x]] \rightarrow \mathbf{No}$ are exactly the surreal infinitesimals, including zero. For general Hahn series we classify the pure monomial prescriptions that admit a homomorphism: the exponent map must be additive and strictly order-reversing, and this condition is sufficient. We also explain the classical exponent-reversing isomorphism, isolate the obstruction inside a countable finitely generated algebra, and supply self-contained

summability arguments and exact finite regression checks.

Provenance. This report is a merge of two independently produced research reports on the same question, circulated as `hahn-evaluation-at-omega` and `reversed-hahn-series`. Both reached the same negative answer by the same argument: each constructed B from the identical quadratic recursion, each verified $B^2 = 1 - x$ coefficient by coefficient, and each applied the hypothetical homomorphism only to that one finite identity. That shared core theorem is therefore proved exactly once below, in Lemma 2.1 and theorem 2.2. The supporting material genuinely diverged, and this document is the union of the two toolkits rather than a deduplication of them: the semiring strengthening of section 3, the general unit square root of Lemma 4.3, the elementary two-support argument of Lemma A.5, and several scope remarks in section 9 come from the second report, while the three-way evaluation criterion, the order-forcing theorem, the monomial classification over an arbitrary exponent group, and the appendix on Higman’s lemma come from the first. Where the two reports phrased an attribution or a priority caveat differently, the more cautious wording has been kept.

Independent research note prepared for Vladimir Reshetnikov. The application to the stated problem is proved here; the Hahn-field background is classical. This manuscript has not undergone independent refereeing or proof-assistant verification.

Contents

1	The problem and its precise scope	2
2	The negative answer	3
3	An obstruction with only nonnegative coefficients	6
4	The universal infinitesimal constraint	7
5	The order hidden in the full Hahn field	11
6	The classical repair: reverse the exponents	12
7	Classification of pure monomial prescriptions	13
8	Why rational evaluation at infinity does not contradict the result	15
9	Verification, attribution, and the boundary of the result	17
10	Conclusion	20
A	Well-ordering and summability from finite words	21
B	Reproducing the artifact checks	24
	References	26

Reading guide. Section 2 contains the complete negative solution and does not depend on the later sections. Section 3 is an independent second obstruction using only nonnegative coefficients and semiring operations; it needs nothing from Section 2 and is not implied by it. Sections 4 and 7 describe exactly what can be evaluated instead. Section 9 collects the scope limits and the misreadings the result invites. Appendix A supplies the order-combinatorial details used only in the extensions. The supplied code checks finite identities; it is not a substitute for the proofs.

1 The problem and its precise scope

1.1 An explicitly posed evaluation question

Lipparini's extended paper [6, Problem 7.7, p. 38] considers formal sums

$$s = \sum_{\beta < \alpha} r_\beta x^{a_\beta}, \quad r_\beta \in \mathbb{R} \setminus \{0\}, \quad \gamma < \beta < \alpha \implies a_\gamma < a_\beta, \quad (1)$$

where α is an ordinal and the exponents are surreal numbers. The question is whether there is a ring homomorphism into **No** satisfying

$$\Phi(x^a) = \omega^a \quad (a \in \mathbf{No}). \quad (2)$$

The support direction is essential: the source exponents increase, whereas Conway normal forms have decreasing exponents. The stated question asks for ordinary ring operations to be preserved, not necessarily infinite sums.

The formula $a = 0$ in (2) forces $\Phi(1) = 1$, regardless of whether one's convention includes unitality in the term "ring homomorphism." The formula $a = 1$ forces $\Phi(x) = \omega$. These are the only two monomial requirements needed for the negative answer.

1.2 Notation and arithmetic conventions

Two warnings are worth stating before any notation. First, the addition and multiplication used throughout this note — in the target and in the exponent group alike — are the surreal *ordered-field* operations, never the noncommutative ordinal operations; $1 - \omega$ below is a negative surreal number, not an attempted ordinal subtraction. Second, we do not assume that Φ fixes every real number. Taking $a = 0$ in (2) already forces $\Phi(1) = 1$, and a unital ring homomorphism necessarily fixes $\mathbb{Q}$. Rational coefficients are all that the obstruction uses, so the status of arbitrary real coefficients never arises in the negative proof.

For a linearly ordered abelian group G , written additively, let

$$\mathcal{H}_\uparrow(G) = \left\{ \sum_{a \in A} r_a x^a : A \subseteq G \text{ is well-ordered, } r_a \in \mathbb{R} \right\}.$$

Zero coefficients are discarded. The zero series has empty support. Each support is a set, including when $G = \mathbf{No}$ is a proper class. Addition is coefficientwise, and multiplication is the Hahn convolution

$$(st)_c = \sum_{a+b=c} s_a t_b. \quad (3)$$

For fixed supports and fixed c , only finitely many summands are nonzero. We give the relevant well-ordering proof in Appendix A. The source of the problem is $\mathcal{H}_\uparrow(\mathbf{No})$.

All exponent addition in (3), when $G = \mathbf{No}$, is surreal addition. All target addition and multiplication are surreal field operations. In particular, $1 - \omega$ is a negative surreal number; it is not an expression in ordinal subtraction. We use ω^a for Conway's omega-map, which satisfies

$$\omega^0 = 1, \quad \omega^{a+b} = \omega^a \omega^b, \quad a < b \implies \omega^a < \omega^b. \quad (4)$$

This notation is not a definition of $\exp(a \log \omega)$ for arbitrary surreal a . No exponential-function compatibility is imposed in this note.

1.3 Which background results are being used?

Conway normal forms identify **No** with real-coefficient generalized series whose surreal exponents are reverse well-ordered. Each such series is a unique surreal number, and its field operations are coefficientwise addition and convolution. The sign is the sign of its leading coefficient. A pure monomial ω^b is infinitesimal exactly when $b < 0$. These facts, together with (4), can be found in the primary treatments [2, pp. 175–176] and [1, Sections 2.3–2.6]; the same conventions are recorded in [3, Sections 1 and 2] and [4].

For the main contradiction, only the much weaker facts that **No** is an ordered field and $\omega > 1$ are required. None of the detailed normal-form theory is needed to construct the obstructing source series.

1.4 What is, and is not, being claimed

The question is still presented as an open problem in the version of the source inspected for this note, and the argument below gives a complete negative answer to that formulation. The square-root construction and the Hahn-field methods themselves are classical; what is claimed here is their application to the specified question, together with an explicit account of the boundary cases. No exhaustive priority determination has been made and this manuscript has not been independently refereed.

The separate problem of constructing useful infinitary operations on arbitrary surreal sequences or on combinatorial games is not settled here. In particular, Lipparini's broader Problem 7.5 is *not* refuted by the failure of this particular ring-homomorphism approach. A partial operation, an operation defined on a smaller domain, or an operation that is not required to be multiplicative can each evade the present obstruction. Section 9 returns to these limits in detail.

2 The negative answer

2.1 Constructing the witness without analytic convergence

An element of $\mathbb{Q}[[x]]$ is a coefficient sequence $(c_n)_{n \geq 0}$, written $\sum_{n \geq 0} c_n x^n$; equality of formal series means equality of every coefficient, and no numerical substitution, convergence, or limiting process is part of the definition. The product of $\sum c_i x^i$ and $\sum d_j x^j$ has $[x^n]$ equal to the *finite* sum $\sum_{i=0}^n c_i d_{n-i}$.

The map

$$\mathbb{Q}[[x]] \longrightarrow \mathcal{H}_\uparrow(\mathbf{No}), \quad \sum_{n \geq 0} c_n x^n \longmapsto \sum_{n \geq 0} c_n x^n, \quad (5)$$

is therefore an *injective ring homomorphism*. Once zero coefficients are discarded, the support is a subset of $\{0, 1, 2, \dots\}$ and hence a well-ordered set of surreal exponents; the Hahn convolution (3) restricts on such supports to precisely the displayed finite Cauchy convolution; and distinct coefficient sequences give distinct supports-with-coefficients. We use (5) silently from now on and regard $\mathbb{Q}[[x]]$ as a subring of $\mathcal{H}_\uparrow(\mathbf{No})$.

Lemma 2.1 (The formal square-root witness). *There is a unique series $B \in \mathbb{Q}[[x]]$ with constant coefficient 1 such that*

$$B^2 = 1 - x. \quad (6)$$

Its support is exactly $\mathbb{N}$, and all its nonconstant coefficients are negative dyadic rationals.

Proof. Define rational coefficients recursively by

$$b_0 = 1, \quad b_1 = -\frac{1}{2}, \quad b_n = -\frac{1}{2} \sum_{k=1}^{n-1} b_k b_{n-k} \quad (n \geq 2), \quad (7)$$

and set $B = \sum_{n \geq 0} b_n x^n$. This is a formal series: it is defined by its coefficient sequence, without a convergence requirement.

The constant coefficient of B^2 is 1, and its coefficient of x is $2b_1 = -1$. For $n \geq 2$ the coefficient of x^n is

$$\sum_{k=0}^n b_k b_{n-k} = 2b_n + \sum_{k=1}^{n-1} b_k b_{n-k} = 0.$$

This proves (6). Read in the other direction, the same computation shows that (7) is forced: with $b_0 = 1$ fixed, requiring each coefficient of B^2 to match that of $1 - x$ determines exactly one new rational number at every stage. The recursion therefore proves existence and uniqueness together.

The recursion preserves dyadic rationals. Moreover, $b_1 < 0$; if $b_1, \dots, b_{n-1}$ are negative, every summand in the recursion for b_n is positive, so $b_n < 0$. Induction proves that none of the coefficients vanishes. $\square$

Theorem 2.2 (Negative answer to Problem 7.7). *There is no ring homomorphism*

$$\Phi : \mathcal{H}_\uparrow(\mathbf{No}) \longrightarrow \mathbf{No}$$

satisfying $\Phi(x^a) = \omega^a$ for every surreal a . More strongly, there is no unital homomorphism $\mathbb{Q}[[x]] \rightarrow F$ into an ordered field F that sends x to an element greater than 1.

Proof. Let u be the image of x and use the series in Lemma 2.1. A unital ring homomorphism would imply

$$\Phi(B)^2 = \Phi(B^2) = \Phi(1 - x) = 1 - u.$$

If $u > 1$, the right-hand side is negative, whereas a square in an ordered field is nonnegative. In the problem, $u = \omega > 1$, and unitality is forced by the prescribed image of x^0 . $\square$

Remark 2.3 (What was not assumed). The proof never writes $\Phi(B) = \sum b_n \Phi(x)^n$. It applies Φ only to the finite algebraic identity $B^2 + x - 1 = 0$. Consequently it does not assume continuity, preservation of a valuation, preservation of order, coefficientwise evaluation, injectivity, or preservation of infinite sums. It also does not assume that arbitrary real coefficients are fixed. Only unitality and the prescribed image of x are used.

2.2 The smallest useful algebraic certificate

The witness can be confined to a countable subring rather than the entire power-series ring or a proper class.

Proposition 2.4. *Let $A = \mathbb{Q}[x, B] \subseteq \mathbb{Q}[[x]]$. Then $A = \mathbb{Q}[B]$ is a countable polynomial algebra in one transcendental element. Nevertheless, no unital homomorphism $A \rightarrow F$ into an ordered field can send its distinguished element x to a number greater than 1.*

Proof. The relation $x = 1 - B^2$ gives $A = \mathbb{Q}[B]$. If B were algebraic over $\mathbb{Q}$, then $x = 1 - B^2$ would be algebraic over $\mathbb{Q}$, contrary to the transcendence of the formal indeterminate x . Thus the map $\mathbb{Q}[Y] \rightarrow A$ sending Y to B is an isomorphism, under which x corresponds to $1 - Y^2$. The obstruction is then just the inequality $1 - y^2 \leq 1$ in every ordered field. Finally, the set of finite rational-coefficient polynomials is countable. $\square$

This does not make A an unembeddable ring: as a polynomial algebra it has many embeddings into ordered fields. It is the requirement on the image of one specified element, not the isomorphism type of the source, that fails. Nor is $\mathbb{Q}[[x]]$ itself countable; only the smaller algebra A is.

The infinite support of B is needed only to *place* the witness inside the proposed domain. The final obstruction is a finite algebraic relation between two elements of a countable ring. Consequently neither uncountable supports nor delicate proper-class considerations enter the negative proof at any point.

2.3 Closed forms for the witness

The coefficients also have the familiar expressions

$$b_0 = 1, \quad b_n = (-1)^n \binom{1/2}{n} = -\frac{\binom{2n}{n}}{4^n(2n-1)} = -\frac{C_{n-1}}{2^{2n-1}} \quad (n \geq 1), \quad (8)$$

where $C_m = \frac{1}{m+1} \binom{2m}{m}$. To check the first expression directly, let $D(x)$ denote the series with coefficients $(-1)^n \binom{1/2}{n}$. Its coefficients obey

$$(n+1)[x^{n+1}]D = (n - \tfrac{1}{2})[x^n]D,$$

so formal differentiation gives $(1-x)D' = -D/2$. For $E = D^2$ this yields $(1-x)E' = -E$, with $E(0) = 1$. If $E = \sum e_n x^n$, then

$$e_1 = -1, \quad (n+1)e_{n+1} = (n-1)e_n \quad (n \geq 1).$$

Thus $E = 1 - x$. The coefficient recursion for a square root with constant coefficient 1 is unique, so $D = B$. The remaining forms in (8) follow by cancelling factorials. This argument is wholly formal; it does not invoke a real analytic branch or a radius of convergence.

A second formal derivation. The same closed form can be verified by a quotient rather than a product, and it is worth recording because the technique is exactly the one reused in Lemma 3.1 below. Write $Q(x) = \sum_{n \geq 0} q_n x^n$ with $q_n = (-1)^n \binom{1/2}{n}$, so that the ratio of consecutive coefficients is

$$2(n+1)q_{n+1} = (2n-1)q_n, \quad \text{equivalently} \quad 2(1-x)Q'(x) + Q(x) = 0.$$

The formal derivative of $Q(x)^2/(1-x)$ is then

$$\frac{2QQ'(1-x) + Q^2}{(1-x)^2} = \frac{Q(2(1-x)Q' + Q)}{(1-x)^2} = 0,$$

where $(1-x)^{-1}$ is the formal geometric series. In characteristic zero a formal series with zero derivative is constant, and here the constant coefficient is 1. Hence $Q^2 = 1 - x$, and the uniqueness clause of Lemma 2.1 identifies Q with B . The two derivations differ in substance and not only in bookkeeping: the first forms $E = D^2$ and reads off a first-order recurrence for its coefficients, whereas the second divides by $1 - x$ and uses that a derivative-free series is constant.

n	$b_n = [x^n](1-x)^{1/2}$	$h_n = [x^n](1-x)^{-1/2}$
0	1	1
1	-1/2	1/2
2	-1/8	3/8
3	-1/16	5/16
4	-5/128	35/128
5	-7/256	63/256
6	-21/1024	231/1024
7	-33/2048	429/2048
8	-429/32768	6435/32768

Table 1: Exact rational coefficients of the two witness series. Both supports have order type ω . The second column is used in sections 2 and 4; the third is the display data for the nonnegative-coefficient witness H of section 3.

3 An obstruction with only nonnegative coefficients

A natural reaction to section 2 is to ask whether the negative coefficients in B are responsible for the contradiction. They are not. Even retaining only nonnegative rational coefficients — so that the source has no additive inverses at all — does not give a multiplicative domain small enough to admit evaluation at ω .

Let $\mathbb{Q}_{\geq 0}[[x]]$ denote the *semiring* of formal power series with nonnegative rational coefficients, under the ordinary coefficientwise addition and the finite Cauchy product recalled at the start of Section 2. Neither operation can leave $\mathbb{Q}_{\geq 0}[[x]]$, so this is a well-defined commutative semiring, and it is not a ring: only the zero series has an additive inverse. A unital semiring homomorphism into an ordered field is required here to preserve only 0, 1, + and $\cdot$. It is not assumed to preserve any order, any infinite sum, or anything else. In this section $G(x) = \sum_{n \geq 0} x^n$ is the formal geometric series, as in (15) and (20); the letter G without an argument continues to denote an exponent group, and the letter H with an argument denotes the witness series about to be defined, never the Hahn field $\mathcal{H}_\uparrow(G)$.

Lemma 3.1 (The nonnegative witness pair). *The series*

$$H(x) = \sum_{n \geq 0} h_n x^n, \quad h_n = \frac{\binom{2n}{n}}{4^n}, \quad G(x) = \sum_{n \geq 0} x^n$$

both lie in $\mathbb{Q}_{\geq 0}[[x]]$ and satisfy

$$H(x)^2 = G(x), \quad G(x) = 1 + xG(x). \quad (9)$$

Proof. Every displayed coefficient is a positive rational, so both series lie in $\mathbb{Q}_{\geq 0}[[x]]$. The second identity in (9) is immediate coefficientwise. For the first, the coefficient ratio

$$2(n+1)h_{n+1} = (2n+1)h_n$$

is a direct computation with central binomial coefficients, and it says exactly that $2(1-x)H'(x) = H(x)$. Calculating now in the containing ring $\mathbb{Q}[[x]]$, the formal derivative of $(1-x)H(x)^2$ is

$$-H^2 + 2(1-x)HH' = -H^2 + H \cdot H = 0.$$

A formal series over a field of characteristic zero with zero derivative is constant, and the constant coefficient of $(1-x)H^2$ is 1. Hence $(1-x)H(x)^2 = 1$. The element $1-x$ is a unit of $\mathbb{Q}[[x]]$ with inverse $G(x)$, by the second identity, so $H(x)^2 = G(x)$.

What has been proved is an equality of nonnegative-coefficient series, even though subtraction was convenient in verifying it. The statement (9) mentions no negative number, and both of its sides lie in $\mathbb{Q}_{\geq 0}[[x]]$. $\square$

Theorem 3.2 (Nonnegative-coefficient obstruction). *Let K be an ordered field and let $u \in K$ satisfy $u \geq 1$. There is no unital semiring homomorphism*

$$\psi : \mathbb{Q}_{\geq 0}[[x]] \longrightarrow K, \quad \psi(x) = u.$$

Proof. Suppose ψ were such a map, and put $z = \psi(H)$ and $g = \psi(G)$. Applying ψ to the two identities of (9) gives

$$g = z^2 \geq 0, \quad g = 1 + ug.$$

The second equation is an equation in the field K , where subtraction is available even though it is unavailable in the source. Rearranging it gives $(1-u)g = 1$. Since $u \geq 1$ and $g \geq 0$, the left-hand side is nonpositive, while the right-hand side is $1 > 0$. This is a contradiction. $\square$

Remark 3.3 (The two obstructions are incomparable). Theorem 3.2 is not a corollary of theorem 2.2, nor conversely, and neither makes the other redundant. It weakens the hypotheses of theorem 2.2 along two axes simultaneously: the source is only a semiring, with no additive inverses and therefore with strictly fewer identities available to the argument, and the excluded values are all $u \geq 1$ rather than only $u > 1$. In the other direction, theorem 2.2 applies to every unital ring homomorphism defined on all of $\mathbb{Q}[[x]]$, a domain on which ψ need not be defined at all, and a ring homomorphism out of $\mathbb{Q}[[x]]$ does not restrict to a semiring homomorphism claim about $\mathbb{Q}_{\geq 0}[[x]]$ unless one already knows that the restriction lands where required. The reader should keep both.

Only the subsemiring generated by x, H, G and the nonnegative rational constants is actually used. In particular, the images of arbitrary positive-coefficient series were never assumed to be positive: the single value that the proof needs to be nonnegative is g , and it is forced to be nonnegative only because it is a square in the target. This is what keeps theorem 3.2 from being a disguised order-preservation hypothesis, and it isolates the algebraic reason that a deliberately nonmonotone interpretation still fails.

4 The universal infinitesimal constraint

The obstruction is not peculiar to ω or to numbers greater than 1. The ring $\mathbb{Q}[[x]]$ forces a much stronger condition on the image of x .

Definition 4.1. In an ordered field F , an element u is *infinitesimal* if

$$|u| < \frac{1}{n} \quad \text{for every integer } n \geq 1.$$

Here zero is included among the infinitesimals. A nonzero infinitesimal may have either sign. When F contains $\mathbb{R}$, this is equivalent to requiring $|u| < r$ for every positive real r , since every positive real exceeds some $1/n$ and every $1/n$ is a positive real.

Lemma 4.2 (Square roots of formal units). *For every $q \in \mathbb{Q}$, the series $1 + qx$ is a square of a unit in $\mathbb{Q}[[x]]$.*

Proof. Substitute $-qx$ for x in $B(x)$: this simply multiplies its coefficient b_n by $(-q)^n$. The resulting formal series $B(-qx)$ satisfies $B(-qx)^2 = 1 + qx$. Its constant coefficient is 1, so it is a unit. Indeed, for a series $v = 1 + \sum_{n \geq 1} v_n x^n$, the inverse coefficients are defined by $w_0 = 1$ and $w_n = -\sum_{k=1}^n v_k w_{n-k}$. $\square$

Lemma 4.2 is all that theorem 4.4 needs, and it has the merit of using nothing beyond the single series B already constructed. The following companion statement is strictly more general in three directions at once — an arbitrary unit rather than $1 + qx$, an arbitrary field of characteristic zero rather than $\mathbb{Q}$, and uniqueness as well as existence — and it is what the arguments of Proposition 8.1 and theorem 4.6 use later. Both are kept: the substitution proof of Lemma 4.2 is the one reproduced in the accompanying one-page extraction, and it does not depend on the recursion below.

Lemma 4.3 (Square roots of units, general form). *Let k be a field of characteristic zero. Every $U \in 1 + xk[[x]]$ has a unique square root in $1 + xk[[x]]$. Moreover, every series in $k[[x]]$ with nonzero constant coefficient is a unit.*

Proof. Write $U = 1 + \sum_{n \geq 1} u_n x^n$ and look for $V = 1 + \sum_{n \geq 1} v_n x^n$ with $V^2 = U$. Comparing the coefficient of x^n on the two sides gives

$$2v_n + \sum_{i=1}^{n-1} v_i v_{n-i} = u_n, \quad \text{that is} \quad v_n = \frac{1}{2} \left(u_n - \sum_{i=1}^{n-1} v_i v_{n-i} \right).$$

Division by 2 is available because $\text{char } k = 0$. The right-hand side involves only $v_1, \dots, v_{n-1}$, so the equation determines exactly one value of v_n at each stage. This is simultaneously the existence and the uniqueness assertion.

For the second claim, let $A = \sum_{n \geq 0} a_n x^n$ with $a_0 \neq 0$ and seek $D = \sum_{n \geq 0} d_n x^n$ with $AD = 1$. Put $d_0 = a_0^{-1}$, and for $n \geq 1$ require the coefficient of x^n in AD to vanish. In that equation the coefficient multiplying the single new unknown d_n is a_0 , which is invertible, so again exactly one value is determined at each stage. $\square$

Theorem 4.4 (Necessary infinitesimal condition). *Every unital ring homomorphism $\phi : \mathbb{Q}[[x]] \rightarrow F$ into an ordered field satisfies*

$$-\frac{1}{n} < \phi(x) < \frac{1}{n} \quad (n \geq 1). \quad (10)$$

In particular, no such homomorphism sends x to a nonzero real number or to a non-infinitesimal element of a non-Archimedean ordered field.

Proof. For each $n \geq 1$, both $1 + nx$ and $1 - nx$ are squares of units by Lemma 4.2. A unital homomorphism maps a unit to a nonzero element: if $vw = 1$, then $\phi(v)\phi(w) = 1$. The images of the two unit squares are therefore strictly positive. With $u = \phi(x)$ this gives

$$1 + nu > 0, \quad 1 - nu > 0,$$

which is exactly (10). $\square$

Proposition 4.5 (The kernel dichotomy). *Let $\phi : \mathbb{Q}[[x]] \rightarrow F$ be a unital homomorphism into a field. If $\phi(x) = 0$, then ϕ is the constant-term map. If $\phi(x) \neq 0$, then ϕ is injective.*

Proof. Unitality forces the rational constants to be fixed. If $\phi(x) = 0$, write $f = f_0 + xg$ to obtain $\phi(f) = f_0$. If $\phi(x) \neq 0$, every nonzero series has a factorization $f = x^m v$ with $m \geq 0$ and v a unit. Both factors have nonzero images, so $\phi(f) \neq 0$. $\square$

Consequently, if the target is Archimedean, the only unital homomorphism $\mathbb{Q}[[x]] \rightarrow F$ is the constant-term map. In particular, every $\mathbb{R}$ -algebra homomorphism $\mathbb{R}[[x]] \rightarrow \mathbb{R}$ is constant-coefficient evaluation: the value of x must be an infinitesimal real number, hence zero, and then the decomposition $f = f(0) + xg$ shows that the value of x determines the whole map. For the Laurent-series field $\mathbb{Q}((x))$ even the constant-term map is unavailable: x is invertible, so its image must be a nonzero infinitesimal. These deductions use only the field axioms and the preceding lemmas.

The corresponding question for a non-Archimedean target needs more care, and the rest of this section is devoted to it. There, nonzero infinitesimal values of x genuinely do occur, so Proposition 4.5 leaves a real alternative open; and the infinite sums that describe the resulting maps must be read formally, through the support calculus of section A, rather than as limits in a topology on the target. We write $\sum^H$ for a sum defined in that way whenever the distinction matters, and section 4.2 shows by an explicit computation that the distinction is not cosmetic.

4.1 Exact converse for surreal targets

For **No** the necessary condition is also sufficient. The relevant notion of infinite sum is Hahn summability, not convergence in an unspecified topology. A family of surreal normal forms is summable when the union of their exponent supports is reverse well-ordered and each exponent occurs in only finitely many members. Such a sum is computed coefficientwise [1, Definition 2.9].

Theorem 4.6 (Exactly which values of x occur). *For a surreal number u , the following are equivalent:*

- (i) *There exists a unital homomorphism $\mathbb{Q}[[x]] \rightarrow \mathbf{No}$ sending x to u .*
- (ii) *u is infinitesimal, including the possibility $u = 0$.*
- (iii) *There exists a coefficient-fixing, strongly additive homomorphism $\mathbb{R}[[x]] \rightarrow \mathbf{No}$ sending x to u .*
- (iv) *There exists an $\mathbb{R}$ -algebra homomorphism $\mathbb{R}[[x]] \rightarrow \mathbf{No}$ sending x to u .*

For nonzero u , a homomorphism in (iii) is given by the Hahn sum

$$\mathrm{ev}_u \left(\sum_{n \geq 0} c_n x^n \right) = \sum_{n \geq 0}^H c_n u^n. \quad (11)$$

It is injective. For $u = 0$, the constant-term map gives the required map.

Proof. The implication (i) $\Rightarrow$ (ii) is theorem 4.4, and (iii) $\Rightarrow$ (i) follows by restriction. Clause (iv) sits between the other two: a coefficient-fixing strongly additive homomorphism is in particular $\mathbb{R}$ -linear, so (iii) $\Rightarrow$ (iv); and an $\mathbb{R}$ -algebra homomorphism is unital, so its restriction to $\mathbb{Q}[[x]]$ witnesses (i) and (iv) $\Rightarrow$ (i). No strong additivity is assumed in (iv). Suppose (ii) holds and $u \neq 0$. All exponents in the normal form of u are strictly negative. Negating this support gives a well-ordered set $A \subseteq \mathbf{No}_{>0}$.

The positive-support summability lemma proved in Appendix A says that the additive monoid generated by A is well-ordered and each exponent has only finitely many representations as a finite word over A . Reflecting exponents back, this shows that the family $(c_n u^n)_{n \geq 0}$ is summable, regardless of the sizes of the real coefficients c_n .

The same finite-representation statement justifies expanding products and regrouping by total degree. Thus, if $f = \sum c_n x^n$ and $g = \sum d_n x^n$,

$$\text{ev}_u(f) \text{ev}_u(g) = \sum_{i,j \geq 0} c_i d_j u^{i+j} = \sum_{n \geq 0} \left(\sum_{i+j=n} c_i d_j \right) u^n = \text{ev}_u(fg).$$

Each coefficient of this regrouping receives only finitely many contributions: a finite representing word has only finitely many places at which it can be split into two words. Addition and preservation of constants are immediate.

For strong additivity, take a source-summable family of power series. At each degree only finitely many members contribute. At a fixed target exponent only finitely many degrees can contribute, by the same positive-support lemma. The two finiteness conditions allow the sums to be interchanged, and their union support lies in a fixed reverse well-ordered set. Hence (11) preserves every source-summable family.

Finally, for $f \neq 0$ write $f = x^m v$, where v is a unit in $\mathbb{R}[[x]]$. The map just constructed sends u^m and v to nonzero elements, so it is injective. The argument for $u = 0$ is the direct constant-term calculation. $\square$

Remark 4.7 (Injectivity again, by leading terms). Injectivity of ev_u also follows from a direct calculation with normal forms, which is worth recording because it produces the leading term of the image rather than merely certifying that it is nonzero, and because it uses no factorization of f and hence neither Lemma 4.3 nor the unit structure of $\mathbb{R}[[x]]$. Let $a < 0$ be the largest exponent occurring in the normal form of the nonzero infinitesimal u , and let $r \neq 0$ be the corresponding leading real coefficient. Let $f = \sum_{n \geq 0} c_n x^n$ be nonzero and let m be the least index with $c_m \neq 0$. The term $c_m u^m$ then has leading exponent ma and leading coefficient $c_m r^m \neq 0$, by (12). For every $n > m$, each exponent occurring in u^n is at most na , and $na < ma$ because $a < 0$ and $n > m$. So no later term of the Hahn sum (11) reaches the exponent ma , and the leading term $c_m r^m \omega^{ma}$ cannot cancel. Hence $\text{ev}_u(f) \neq 0$.

Strong additivity makes (11) the unique coefficient-fixing strongly additive map with that value of x , because every power series is a summable family of its monomials. We do *not* infer uniqueness among arbitrary abstract ring homomorphisms. Describing the maps that fail to respect infinite sums is a separate question, and the distinction between arbitrary and strongly additive maps is a substantive one in the surreal setting; compare [3, Sections 3 and 4]. The successful termwise construction is the familiar change-of-value-group construction for Hahn fields, written in the opposite exponent convention; compare [3, Section 2.3]. In particular, the nonexistence theorem requires far less structure than the positive construction uses.

There is no contradiction with classical analytic convergence. A series may have zero radius of convergence over the real numbers and still admit (11): its real coefficients change the values at exponents, but they do not change the well-ordering or finite-occurrence condition on supports.

4.2 Summability is not convergence in the full order topology

The sums above are defined by the support calculus of section A, not as limits. That is not a matter of taste. Consider the entirely legitimate Hahn sum

$$s = \sum_{n \geq 0}^H \omega^{-n} = \frac{1}{1 - \omega^{-1}},$$

which is (15) again, and let $s_N = \sum_{n=0}^N \omega^{-n}$ be its N -th partial sum, a finite and therefore unambiguous surreal number. Summing the remaining geometric tail gives

$$s - s_N = \sum_{n \geq N+1}^H \omega^{-n} = \frac{\omega^{-(N+1)}}{1 - \omega^{-1}} > \omega^{-\omega} \quad (N \geq 0),$$

the last inequality because $0 < 1 - \omega^{-1} < 1$ forces $(1 - \omega^{-1})^{-1} > 1$, while $-(N+1) > -\omega$ for every natural number N and the omega-map is strictly increasing by (4). The single fixed positive surreal $\omega^{-\omega}$ therefore determines a neighbourhood of s in the order topology that *no* partial sum s_N ever enters. The family is summable and its sum is s , yet the sequence of partial sums does not converge to s in any order-theoretic sense. This is why theorem 4.6 is stated in terms of Hahn summability and why no appeal to convergence is made anywhere in this note.

5 The order hidden in the full Hahn field

The main proof deliberately avoids assuming source-order preservation. In the full Hahn field, however, order preservation is not an optional extra: it can be recovered from squares.

5.1 Leading terms and the canonical order

For $s \neq 0$ in $\mathcal{H}_\uparrow(G)$ let

$$v(s) = \min \text{supp}(s), \quad \text{lc}(s) = s_{v(s)}.$$

Declare $s > 0$ if $\text{lc}(s) > 0$. For two nonzero series,

$$v(st) = v(s) + v(t), \quad \text{lc}(st) = \text{lc}(s) \text{lc}(t). \quad (12)$$

Indeed, the smallest possible exponent of a product is the sum of the two smallest exponents; it is attained only by that pair. The coefficient cannot cancel. The sum of two positive series is positive as well: at the smaller leading exponent its coefficient is positive, or, when the exponents agree, it is the sum of two positive coefficients. Thus this is a compatible total ordering.

For $a > 0$, the monomial x^a is positive, and for every positive real r the series $r - x^a$ has leading coefficient r at exponent 0. Therefore

$$0 < x^a < r \quad (a > 0, r \in \mathbb{R}_{>0}). \quad (13)$$

Larger exponents encode smaller magnitudes in this source convention.

Lemma 5.1 (Every canonically positive element is a square). *Suppose G is 2-divisible: every $a \in G$ has a half $a/2 \in G$. Then the positive elements of $\mathcal{H}_\uparrow(G)$ are exactly its nonzero squares.*

Proof. A nonzero square has positive leading coefficient by (12). Conversely, if $s > 0$, factor its leading term:

$$s = rx^a(1 + \epsilon), \quad r > 0, \quad \text{supp}(\epsilon) \subseteq G_{>0}.$$

By the summability lemma in Appendix A,

$$q = \sqrt{r} x^{a/2} \sum_{n \geq 0} \binom{1/2}{n} \epsilon^n$$

is a well-defined Hahn series. Formal substitution of ϵ into the power-series square-root identity is legitimate coefficientwise, again by finite representability. It gives $q^2 = s$. $\square$

Theorem 5.2 (Order is algebraically forced). *For 2-divisible G , the field $\mathcal{H}_\uparrow(G)$ has only one ordering compatible with its field operations. Every unital ring homomorphism from $\mathcal{H}_\uparrow(G)$ into an ordered field preserves that ordering.*

Proof. In any ordered field every nonzero square is positive. By Lemma 5.1, each nonzero element is either a square or the negative of a square according to the canonical order. Hence every compatible ordering has the same positive cone.

A unital homomorphism out of a field is injective: if $s \neq 0$, then $\phi(s)\phi(s^{-1}) = 1$. A positive source difference is a nonzero square, so its image is a nonzero square and is positive in the target. This is exactly order preservation. $\square$

The fact that $\mathcal{H}_\uparrow(G)$ is a field follows from the formal geometric inverse. Concretely, writing a nonzero s as $s = rx^a(1 + \epsilon)$ with $r \neq 0$, $a = v(s)$ and $\text{supp}(\epsilon) \subseteq G_{>0}$, the inverse is

$$s^{-1} = r^{-1} x^{-a} \sum_{n \geq 0} (-\epsilon)^n,$$

a sum that is legitimate by the support calculus rather than by convergence. The full proof, valid for an arbitrary linearly ordered abelian group G , is Proposition A.7 in Appendix A; it is stated there in that generality rather than only for $G = \mathbf{No}$. No real-closedness theorem is needed for theorem 5.2, and having all positive elements square should not, by itself, be confused with real closedness.

Applied to $G = \mathbf{No}$, this gives a second explanation of the impossibility: x is a positive infinitesimal in the source, but the requested image is positive infinite. One cannot rescue the prescription by asking for a non-monotone ring homomorphism, since there is no such freedom on this field. The short proof in Section 2 is nevertheless stronger as a local obstruction: it works before adjoining any fractional exponents and without proving uniqueness of the source ordering.

6 The classical repair: reverse the exponents

The source field is not intrinsically incompatible with the surreals. The correct normal-form identification reverses the exponent order. This is a classical construction: van den Dries and Ehrlich explicitly display its restriction to any set-sized surreal exponent group on p. 176 of van den Dries and Ehrlich [2].

Theorem 6.1 (Exponent-reversing isomorphism). *The map*

$$J : \mathcal{H}_\uparrow(\mathbf{No}) \longrightarrow \mathbf{No}, \quad J\left(\sum_{a \in A} r_a x^a\right) = \sum_{a \in A} r_a \omega^{-a} \quad (14)$$

is a coefficient-fixing ordered-field isomorphism. It is strongly additive and satisfies $J(x) = \omega^{-1}$.

Proof. If A is well-ordered, $-A$ is reverse well-ordered. The expression on the right of (14) is therefore a legitimate Conway normal form and so defines a unique surreal number. Negation of exponents is injective, hence there is no identification or cancellation between distinct source monomials.

The map preserves addition coefficientwise. For multiplication, the coefficient at target exponent $-c$ is the finite sum over pairs satisfying $-a - b = -c$, equivalently $a + b = c$. This is exactly the source convolution. It preserves 1 and fixes every real coefficient.

A nonzero source leading term rx^a becomes the target leading term $r\omega^{-a}$, with the same coefficient, so signs are preserved. Finally, any surreal normal form $\sum_b c_b \omega^b$ has the inverse image $\sum_b c_b x^{-b}$; the reflected support is well-ordered. This proves surjectivity as well as injectivity. For a summable family, reflection preserves the finite-occurrence property and changes the union support from well-ordered to reverse well-ordered. Thus J is strongly additive. $\square$

6.1 Two explicit evaluations

For the formal geometric series $G(x) = \sum_{n \geq 0} x^n$ we have

$$J(G) = \sum_{n \geq 0} \omega^{-n} = \frac{1}{1 - \omega^{-1}}. \quad (15)$$

For the obstructing square-root series,

$$J(B) = 1 - \frac{\omega^{-1}}{2} - \frac{\omega^{-2}}{8} - \frac{\omega^{-3}}{16} - \cdots, \quad J(B)^2 = 1 - \omega^{-1}. \quad (16)$$

The right side of the square identity is positive. Its displayed root has positive leading coefficient and is therefore the positive square root. There is no order conflict.

6.2 Exponent reversal is not field-order reversal

The word “reversing” in theorem 6.1 applies to the exponent parameter, not to the ordering of field elements. The map $a \mapsto -a$ reverses the exponent order, while the map J preserves field order. This is exactly what is needed: source positive exponents mean small monomials, whereas positive exponents in ω^a mean large monomials.

Replacing x by ω^{-1} in this way is a repair of the prescription, not a positive solution of Problem 7.7. Its demanded images are ω^a , not ω^{-a} .

7 Classification of pure monomial prescriptions

We can classify the monomial data compatible with a homomorphism without assuming in advance that the homomorphism is strongly additive.

Definition 7.1. A map between Hahn-series structures is *strongly additive* if it takes every summable family to a summable family and commutes with its sum. For $\mathcal{H}_\uparrow(G)$, summability means that the union of exponent supports is well-ordered and each exponent occurs in only finitely many family members. For $\mathbf{No}$, the union support is instead reverse well-ordered.

Theorem 7.2 (Complete existence criterion for pure monomial data). *Let G be a set-sized linearly ordered abelian group, or let $G = \mathbf{No}$ with set-sized supports. Let $f : G \rightarrow \mathbf{No}$ be a function. The following conditions are equivalent:*

- (i) *There exists a unital ring homomorphism $\Phi : \mathcal{H}_\uparrow(G) \rightarrow \mathbf{No}$ such that $\Phi(x^a) = \omega^{f(a)}$ for every $a \in G$.*
- (ii) *The map f is additive and strictly order-reversing:*

$$f(a + b) = f(a) + f(b), \quad a < b \implies f(a) > f(b). \quad (17)$$

- (iii) *There exists a coefficient-fixing, strongly additive homomorphism with those monomial images.*

When these conditions hold, the map in (iii) is unique and is the embedding

$$T_f \left(\sum_{a \in A} r_a x^a \right) = \sum_{a \in A} r_a \omega^{f(a)}. \quad (18)$$

Moreover, T_f is onto $\mathbf{No}$ if and only if f is onto $\mathbf{No}$.

Proof. Necessity of additivity. Unitality and the identity $x^{a+b} = x^a x^b$ give

$$\omega^{f(0)} = 1 = \omega^0, \quad \omega^{f(a+b)} = \omega^{f(a)} \omega^{f(b)} = \omega^{f(a)+f(b)}.$$

Injectivity of the omega-map implies $f(0) = 0$ and additivity.

Necessity of order reversal. Fix $a > 0$. Substitution of the monomial x^a for a new formal variable z embeds $\mathbb{Q}[[z]]$ into $\mathcal{H}_\uparrow(G)$:

$$\sum_{n \geq 0} q_n z^n \mapsto \sum_{n \geq 0} q_n x^{na}.$$

The support is well-ordered because $0, a, 2a, \dots$ is strictly increasing; convolution agrees with ordinary power-series multiplication. Compose this embedding with Φ . By theorem 4.4, its image of z , namely $\omega^{f(a)}$, is infinitesimal. A pure Conway monomial is a positive infinitesimal exactly when its exponent is negative. Thus $f(a) < 0$ for all $a > 0$. If $a < b$, additivity gives

$$f(b) - f(a) = f(b - a) < 0,$$

which is strict order reversal. This argument does not require G to be divisible or Φ to fix the real coefficients.

Sufficiency and construction. Assume (ii). The image $f(A)$ of every well-ordered support is reverse well-ordered, and f is injective. Therefore (18) is a valid normal form. Addition is preserved coefficientwise. For products, additivity identifies $f(a) + f(b)$ with $f(a + b)$, and injectivity ensures that equality of image exponents is equivalent to equality of their source sums. Hence the finite convolution coefficients agree.

The construction fixes the real constants and preserves 1. A nonzero coefficient remains nonzero at a unique image exponent, so T_f is injective. A summable family has a well-ordered union support

and finite occurrence at each source exponent. Its image has a reverse well-ordered union support and the identical occurrence counts. This proves strong additivity and (iii). The implication (iii) $\Rightarrow$ (i) is immediate.

Uniqueness under strong additivity. Every source series is the sum of its summable family of coefficient-monomial terms. A coefficient-fixing strongly additive map must therefore have exactly the value in (18).

Surjectivity. If f is onto, apply f^{-1} to the reverse well-ordered support of an arbitrary surreal normal form; the resulting support is well-ordered and provides a preimage. Conversely, if T_f is onto, every monomial ω^b is in its image. Normal-form uniqueness in (18) forces $b = f(a)$ for a source exponent a . $\square$

The equivalence of (i) and (ii) is an existence classification even for ordinary, potentially non-strong homomorphisms. The uniqueness assertion, by contrast, applies only to the coefficient-fixing strongly additive maps. These are distinct statements and should not be conflated.

7.1 Examples and boundary cases

Example 7.3 (A family of isomorphisms). For any fixed $c \in \mathbf{No}_{>0}$, set $f_c(a) = -ca$, where multiplication is surreal field multiplication. This map is additive, strictly order-reversing, and bijective on $\mathbf{No}$. Thus

$$T_c \left(\sum_a r_a x^a \right) = \sum_a r_a \omega^{-ca} \quad (19)$$

is a coefficient-fixing ordered-field isomorphism $\mathcal{H}_\uparrow(\mathbf{No}) \rightarrow \mathbf{No}$. Its inverse is obtained by replacing an exponent b by $-b/c$. In particular, x goes to ω^{-c} , a positive infinitesimal even when c itself is infinitesimal.

Example 7.4 (The forbidden identity exponent map). The prescription in Problem 7.7 is $f(a) = a$. It is additive but order-preserving, not order-reversing. It fails the criterion on the single positive exponent $a = 1$. Thus the general classification recovers the short contradiction.

Example 7.5 (Zero is not a Hahn-field monomial image). In theorem 4.6, sending x to zero is allowed because x is not a unit of $\mathbb{Q}[[x]]$. In $\mathcal{H}_\uparrow(G)$ every monomial is a unit, with inverse x^{-a} . It cannot have image zero under a unital homomorphism. This accounts for an important difference between the power-series ring and a full Hahn field.

An increasing exponent sequence can have arbitrary ordinal length, but an order-reversing f handles all such supports simultaneously. No bound on their ordinal types or on the birthday of their surreal exponents is needed. That is the order-theoretic content of the sufficient direction.

8 Why rational evaluation at infinity does not contradict the result

The failure is not that every formal expression becomes meaningless at ω . Large useful subfields admit precisely that evaluation. What fails is extending it to the entire power-series or Hahn field.

8.1 The rational subfield

Proposition 8.1. *The assignment $x \mapsto \omega$ extends to an injective, real-coefficient-fixing homomorphism*

$$\text{ev}_\omega : \mathbb{R}(x) \longrightarrow \mathbf{No}, \quad \frac{p(x)}{q(x)} \longmapsto \frac{p(\omega)}{q(\omega)}.$$

Here $\mathbb{R}(x)$ is regarded as a subfield of $\mathbb{R}((x))$ by its Laurent expansion at zero.

Proof. First, the Laurent embedding is genuinely available and not merely asserted. Let $q(x)$ be a nonzero polynomial and factor it as $q(x) = x^m q_0(x)$ with $q_0(0) \neq 0$. By the second clause of Lemma 4.3, q_0 is a unit of $\mathbb{R}[[x]]$, so $1/q = x^{-m} q_0^{-1}$ lies in $\mathbb{R}((x))$. Hence every rational function has a formal Laurent expansion at zero, and $\mathbb{R}(x) \subseteq \mathbb{R}((x))$ as claimed.

For the evaluation, let $p(X) = a_d X^d + \cdots + a_0$ be a nonzero polynomial with $a_d \neq 0$. The surreal number $p(\omega)$ has leading normal-form term $a_d \omega^d$, which dominates the finitely many lower monomials in absolute value. It is therefore nonzero. Thus ω is transcendental over the real coefficient field, and substitution extends from the polynomial ring to its fraction field. Well-definedness follows by cross multiplication and injectivity follows from the same nonvanishing statement. $\square$

The element $G(x) = 1 + x + x^2 + \cdots$ belongs to this rational subfield because

$$(1 - x)G(x) = 1.$$

Hence the rational evaluation does consistently assign it

$$\text{ev}_\omega(G) = \frac{1}{1 - \omega} = - \sum_{n \geq 1} \omega^{-n} < 0. \quad (20)$$

The last series is a legitimate decreasing-exponent normal form. In particular, the geometric-series calculation by itself is not a contradiction to the existence of an ordinary ring homomorphism. Positivity of that source series need not be preserved on the rational subfield. That field carries more than one compatible field ordering: there is one in which x is a positive infinitesimal, namely the one induced from $\mathbb{R}((x))$ by leading terms, and another in which x is positive and infinite, namely the one pulled back along $x \mapsto \omega$. An algebraic embedding is free to switch between two such orderings, and ev_ω does exactly that. What cannot be transported by any embedding is a *square root* of a source element whose image is negative, since a square stays a square.

The new obstruction appears upon adjoining B : the compatible image would have to solve $Y^2 = 1 - \omega$. Thus Proposition 8.1 cannot extend to $\mathbb{R}(x, B)$, although it is perfectly valid on $\mathbb{R}(x)$.

8.2 A contrasting quadratic extension

Let $C(x) = B(-x) \in \mathbb{R}[[x]]$, so that $C(x)^2 = 1 + x$ and $C(0) = 1$. Unlike B , this series can be adjoined to the rational domain of evaluation. The polynomial $Y^2 - (1 + x)$ is irreducible over $\mathbb{R}(x)$: the rational function $1 + x$ has a zero of odd multiplicity at $x = -1$, whereas a square of a rational function has even multiplicity at every zero and pole.

Since $1 + \omega > 0$, it has a positive square root in $\mathbf{No}$. Concretely,

$$\sqrt{1 + \omega} = \omega^{1/2} \sum_{n \geq 0} \binom{1/2}{n} \omega^{-n},$$

which is a valid normal form with square $1 + \omega$. Thus $x \mapsto \omega$ and $C \mapsto \sqrt{1 + \omega}$ define an embedding of $\mathbb{R}(x, C)$ into **No**.

This embedding does not evaluate the original Taylor series of C term by term at ω . Instead it respects the finite algebraic relation and uses a different, decreasing-exponent expansion of its chosen value. Moreover, either sign of the square root gives a field embedding, illustrating why algebraic extensions may involve choices even when x is fixed.

Proposition 8.2 (Criterion for a simple algebraic extension). *Let $p(Y) \in \mathbb{R}(x)[Y]$ be monic and irreducible, and let $L = \mathbb{R}(x)[Y]/(p)$. The evaluation $x \mapsto \omega$ extends to an embedding $L \rightarrow \mathbf{No}$ if and only if the polynomial $p^\omega(Y)$ obtained by evaluating its rational-function coefficients at ω has a root in **No**.*

Proof. Any extension sends the residue class of Y to such a root. Conversely, a root induces a unital homomorphism from $\mathbb{R}(x)[Y]$ to **No** that annihilates (p) . It factors through L , which is a field because p is irreducible. A unital homomorphism from a field has zero kernel, so the induced map is an embedding. $\square$

The pair $p = Y^2 - (1 - x)$ and $p = Y^2 - (1 + x)$ therefore separates two very similar formal Taylor series: the first algebraic extension is forbidden at positive infinity, while the second is permitted. This is a field-theoretic obstruction, not merely a rule about rearranging divergent series.

It should not be inferred that $\mathbb{R}(x)$ is a *largest* subfield on which evaluation at ω is admissible, and no such claim is made here. The preceding paragraphs exhibit a proper algebraic extension $\mathbb{R}(x, C)$ of it on which the evaluation still exists, precisely because $1 + \omega$ is positive and so has a square root in **No**. What theorem 2.2 identifies is one particular forbidden algebraic extension, not every algebraic extension; we do not classify the maximal admissible subfields.

9 Verification, attribution, and the boundary of the result

9.1 What has been proved

Theorem 2.2 gives a complete negative answer to the precise ring homomorphism question in Problem 7.7. Its proof is an explicit algebraic contradiction, not a computational search with no examples found. It does not settle the separate questions in the surrounding paper about other possible infinitary operations on surreal numbers or combinatorial games.

The additional conclusions are exact statements with distinct scopes: Theorem 3.2 strengthens the obstruction to a semiring source with no additive inverses and to all values $u \geq 1$; Theorem 4.6 classifies possible values of one power-series variable in **No**; Theorem 7.2 classifies pure monomial data for Hahn-field homomorphisms; Theorem 6.1 recalls the standard exponent-reversing identification. We do not classify all arbitrary embeddings of the surreal field, all non-monomial Hahn substitutions, or all maximal subfields on which evaluation at ω is possible.

Stated as a ledger, exactly four things are proved: the nonexistence theorem for the posed prescription; its nonnegative-coefficient semiring strengthening; the image criterion for a single power-series variable; and the existence criterion for prescribed monomial images. Everything else in this note is either classical background, a worked example, or a scope disclaimer.

9.2 The normalization cannot be omitted

If one merely asks for *some* unital ring map $\mathcal{H}_\uparrow(\mathbf{No}) \rightarrow \mathbf{No}$, one exists: theorem 6.1 supplies an isomorphism. The condition $\Phi(x^a) = \omega^a$ is therefore essential to the problem and not a piece of decoration on it. In the other direction, the condition is used only very weakly: the much smaller pair of requirements $\Phi(1) = 1$ and $\Phi(x) = \omega$ already contradicts the subring certificate of Proposition 2.4, so the values at all other monomials are never consulted. Finally, dropping unitality does admit a map, namely the zero map, but that map fails the monomial condition already at $a = 0$, where it gives 0 rather than $\omega^0 = 1$. There is no gap between these three observations for a counterexample to hide in.

9.3 Finite supports, polynomials, and truncations

The ring of *finite* formal sums of monomials can be evaluated termwise at ω , since no infinite support needs interpretation; $\mathbb{R}[x]$ and the rational domain of Proposition 8.1 present no obstruction whatever. It is instructive that B is absent from the polynomial ring for a reason that has nothing to do with the target: a square polynomial has even degree, whereas $1 - x$ has degree one. The question thus fails only once the domain is enlarged far enough to contain the specific algebraic power series B , and not merely because formal monomials with infinite supports are permitted.

Nor can the contradiction be reached, or evaded, by evaluating truncations. For $B_N = \sum_{n \leq N} b_n x^n$ the identity $B_N^2 = 1 - x$ is simply *false*; it holds only modulo x^{N+1} . At $x = \omega$ the discarded high-degree terms are not small — they are the dominant ones. The contradiction in theorem 2.2 is about the existence of an exact homomorphism on the series ring, not about a limiting calculation with polynomials, and a reader who reconstructs it as the latter will find it unconvincing for the right reason.

9.4 All support lengths are set-sized

Although the exponents range over a proper class, each individual series in (1) has ordinal-indexed and therefore set-sized support. No sum indexed by the entire class of ordinals is introduced anywhere. The positive-support arguments of Appendix A use only sets of exponents and finite words over those sets, and the obstruction itself lives entirely inside the set-sized ring $\mathbb{Q}[[x]]$ together with a hypothetical image in an ordered field. Class-sized notation, where it appears, can be handled in a standard class theory such as NBG. Consequently the negative result cannot be avoided by changing the treatment of proper classes.

9.5 The target must be formally real for this obstruction

The argument uses exactly one property of $\mathbf{No}$ beyond its ring structure: that a square is nonnegative in an ordered field. It does not claim nonexistence for every imaginable target ring. If a target contains an element whose square is $1 - u$, this particular certificate produces no contradiction there. Allowing non-real values would of course change the problem, which is posed with a surreal-valued map; no assertion about a global homomorphism into a complexified target is made or implied here.

9.6 Classical ingredients versus the application

The formal binomial square root, the leading-term order on Hahn fields, positive-support summability, the omega-map, and exponent reversal are classical. In particular, (14) is not being proposed as a new isomorphism; its set-sized versions occur explicitly in van den Dries and Ehrlich [2, p. 176]. The word-embedding result used in the appendix is Higman's classical theorem [5]; we give the special-case proof needed here.

The contribution of this note is the direct application of the square-root identity to the specifically posed evaluation question and the organized account of its exact consequences. The elementary nature of the obstruction is a reason to make its assumptions unusually explicit, not a reason to claim a new general theory of Hahn fields.

9.7 Source version and priority qualification

The inspected arXiv record lists the revised version dated 30 April 2026 as its latest version. The question is still explicit there as Problem 7.7; the earlier formulation is Problem 6.8 in the May 2025 version. Numbering in this note refers to the extended preprint, not to an assumption that the question occurs in the shortened journal version [6].

A focused public-source search was carried out on 20 September 2026 for subsequent resolutions and for the specific Hahn/surreal homomorphism formulation. No prior resolution was located in that search. This is not an exhaustive priority certification, and it cannot rule out an unpublished observation or an unindexed later discussion. The proof stands independently of that bibliographic uncertainty. The manuscript has not been independently refereed or checked in a proof assistant.

9.8 Exact finite regression checks

The archive includes `code/verify.py`, using only Python's standard library and exact `Fractions.Fraction` arithmetic. No floating-point number is created anywhere in it. The executed default run verified the square identity through degree 256, that is,

$$B_{256}(x)^2 \equiv 1 - x \pmod{x^{257}},$$

and the two semiring identities of Lemma 3.1 to the same modulus,

$$H(x)^2 \equiv G(x), \quad B(x)H(x) \equiv 1, \quad (1 - x)G(x) \equiv 1 \pmod{x^{257}}.$$

It compared 257 coefficients of B obtained independently from five routes: the quadratic recursion of Lemma 2.1, the general unit square-root recursion of Lemma 4.3 applied to $1 - x$, the first-order binomial recursion, the Catalan closed formula, and Catalan numbers regenerated from the convolution recurrence $C_m = \sum_i C_i C_{m-1-i}$. It verified that all computed nonconstant coefficients of B are negative and dyadic, that all 257 computed coefficients of H are strictly positive, and that both differential recurrences $2(n+1)b_{n+1} = (2n-1)b_n$ and $2(n+1)h_{n+1} = (2n+1)h_n$ hold at every checked degree.

For unit square roots there are two independent batteries at degree 64: the 32 scaled checks for $1 \pm nx$ with $1 \leq n \leq 16$, obtained by scaling the coefficients of B , and 12 checks for $1 + cx$ at the signed rational values $c \in \{\pm 1, \pm 2, \pm 3, \pm \frac{1}{2}, \pm \frac{2}{3}, \pm \frac{7}{5}\}$, obtained from the general routine. The rational values exercise code that the integer scalings do not.

For exponent reversal there are likewise two harnesses, and both are run. The first performs 1000 deterministic randomized tests of finite sparse rational-exponent polynomials at randomly drawn positive rational scales, with seed 20260920. The second performs 200 polynomial pairs at each of the three fixed scales 1, 2 and $\frac{3}{2}$, that is 600 scale-pair cases, with seed 1729; the non-integer fixed scale is not sampled by the first harness in the same way. Each case checks addition, multiplication, and strict reversal of the support order under $a \mapsto -ca$. The output is recorded in `results/verification.json`; all 257 exact coefficients of both B and H , together with the coefficients of B^2 and H^2 , are in `results/coefficients.csv`.

These computations do not verify infinite summability, the surreal number axioms, or the universal quantifiers of the theorems. The all-degrees proofs are the coefficient recursion in Lemma 2.1 and the derivative argument in Lemma 3.1. The checks instead detect sign, indexing, coefficient, and finite-convolution errors in the concrete formulas and implementation.

9.9 A compact logical audit

The contradiction uses an element that really belongs to the proposed source: its support is the allowed increasing sequence $0, 1, 2, \dots$, and its coefficients are nonzero real numbers. The identity is proved in that source using its own multiplication. Unitality follows from the prescribed image of x^0 . The target inequality is a field-order statement, not an ordinal calculation. No image is assigned to an infinite sum by distributing Φ over its terms. These checks isolate all assumptions needed for the answer.

9.10 What the result does not settle

The conclusions above are specific to *multiplicative* evaluations on the stated series domains. They do not prohibit useful regularizations of selected divergent series, they say nothing about nonmultiplicative summation conventions, and they do not constrain game-valued operations formulated with different axioms. They likewise do not constitute a classification of arbitrary surreal field embeddings. As recorded in section 1.4, Lipparini’s broader Problem 7.5 remains open as far as this note is concerned: a partial operation, an operation on a smaller domain, or an operation not required to be multiplicative evades everything proved here.

10 Conclusion

The requested extension cannot exist. The full obstruction is already

$$B(x)^2 = 1 - x \quad \text{in } \mathbb{Q}[[x]], \quad 1 - \omega < 0 \quad \text{in } \mathbf{No}.$$

The increasing-support convention carries an algebraic commitment: its positive exponent monomials behave as infinitesimals in any ordered-field image. This commitment can be read from square roots of units even when no order-preservation axiom is imposed.

The obstruction is already countable, rational and quadratic. It persists without continuity, without order preservation, without any hypothesis about infinite sums, and — by theorem 3.2 — even in a nonnegative-coefficient semiring formulation in which subtraction is unavailable in the source and the excluded values are widened from $u > 1$ to $u \geq 1$.

There is nevertheless an exact positive picture. Ordinary power series can be evaluated at every surreal infinitesimal, and no other surreal values of the variable are possible. General pure

monomial evaluations exist precisely for additive, strictly order-reversing exponent maps. The canonical example is $x^a \mapsto \omega^{-a}$. Thus the failure at ω is not a shortage of surreal values or a class-size difficulty: it is a reversal of the magnitude orientation required by the source algebra.

A Well-ordering and summability from finite words

This appendix proves the support facts used in the extensions. It is not needed for the main contradiction, whose coefficients have finite ordinary Cauchy convolutions. We work in the usual set-theoretic setting with choice. Every support below is a set, even if the ambient ordered group is a class.

A.1 Well-ordered sets and finite products

A set in a linear order is *well ordered* when every nonempty subset has a least element, and *reverse well ordered* when every nonempty subset has a greatest element. The two notions are not interchangeable: $\{0, 1, 2, \dots\}$ is well ordered and $\{0, -1, -2, \dots\}$ is reverse well ordered, and neither set has the other property. The whole orientation question of this note is the difference between them.

Lemma A.1 (Nondecreasing subsequences). *Every infinite sequence in a well-ordered set has an infinite nondecreasing subsequence. Consequently, every infinite sequence in a finite product of well-ordered sets has an infinite coordinatewise nondecreasing subsequence.*

Proof. For the first claim, choose an occurrence of the least value taken by the whole sequence, then an occurrence of the least value taken on the remaining infinite tail, and continue. The successive tail minima cannot decrease, since each is the minimum of a subset of the previous tail.

For the second, apply the first claim to the first coordinate to obtain an infinite subsequence nondecreasing there, then to the second coordinate of that subsequence, and so on. A finite product requires only finitely many extractions, and each extraction preserves the monotonicity already achieved. $\square$

A.2 Finite words over a well-ordered alphabet

Let A be a well-ordered set. For finite words $u = (u_1, \dots, u_m)$ and $v = (v_1, \dots, v_n)$ over A , write $u \leq v$ when there are indices

$$1 \leq i_1 < \dots < i_m \leq n \quad \text{such that} \quad u_j \leq v_{i_j} \quad (1 \leq j \leq m).$$

The empty word embeds into every word. A sequence is called *bad* if no earlier member embeds into a later one.

Lemma A.2 (Higman's lemma, well-ordered alphabet case). *Every infinite sequence of finite words over A contains an increasing pair for $\leq$.*

Proof. Suppose a bad infinite sequence exists. Construct a bad sequence $w_0, w_1, \dots$ with the following minimality property: having chosen $w_0, \dots, w_{n-1}$, choose w_n to have the smallest possible length among words permitting that prefix to be extended to a bad infinite sequence. The possible lengths form a nonempty subset of $\mathbb{N}$, so a least one exists at every stage.

No w_n is empty, since an empty word embeds into every later word. Write $w_n = a_n v_n$, with first letter $a_n \in A$ and a shorter tail v_n . By the first part of Lemma A.1, choose

$$i_0 < i_1 < i_2 < \cdots, \quad a_{i_0} \leq a_{i_1} \leq a_{i_2} \leq \cdots.$$

Consider the modified sequence

$$w_0, \dots, w_{i_0-1}, v_{i_0}, v_{i_1}, v_{i_2}, \dots$$

It is bad. An embedding between two words in the unchanged prefix would already contradict badness of the original sequence. If a prefix word embeds into a tail v_{i_j} , it embeds into w_{i_j} as well, again a contradiction. Finally, if $v_{i_j} \leq v_{i_k}$ for $j < k$, adjoining the first letters with $a_{i_j} \leq a_{i_k}$ gives $w_{i_j} \leq w_{i_k}$, also impossible.

The modified sequence therefore extends the prefix through index $i_0 - 1$ to a bad sequence using the strictly shorter word v_{i_0} at the next position. This contradicts the chosen minimality of w_{i_0} . $\square$

This is the standard minimal-bad-sequence mechanism for the special case of Higman's theorem [5] used below; no general tree theorem is needed.

A.3 The positive-support lemma

Lemma A.3 (Positive-support summability). *Let G be a linearly ordered abelian group, and let $A \subseteq G_{>0}$ be well-ordered. Let $M(A)$ be the set of sums of finite words over A , including the empty sum 0. Then:*

- (a) $M(A)$ is well-ordered.
- (b) For every $g \in G$, only finitely many finite words over A have sum g .

In (b) it makes no difference whether words are counted as nondecreasing words or as arbitrary ordered finite tuples: a fixed finite word has only finitely many permutations, so the two counts are finite together.

Proof. If $u \leq v$, the sum of the letters of u is at most the sum of the letters of v . If $u \neq v$, this inequality is strict: either there is an extra positive letter in v , or one of the matched letters is strictly larger. When the word lengths agree, the only strictly increasing bijection of their index sets is the identity, so the latter alternative exhausts the distinct-word case.

If $M(A)$ contained an infinite strictly decreasing sequence of sums, choose one representing word for each sum. An increasing pair of words provided by Lemma A.2 would give a nondecreasing pair of sums, a contradiction. A linearly ordered set with no infinite strictly decreasing sequence is well-ordered in the present set-theoretic setting: from a nonempty subset with no minimum one could choose a descending sequence. This proves (a).

If infinitely many distinct words had the same sum g , enumerate a countable sequence of distinct such words. An increasing pair from Lemma A.2 would have strictly increasing sums by the preceding observation, again a contradiction. This proves (b). $\square$

Notice that no Archimedean property is assumed. The positive elements of A may themselves be infinitesimal in a larger ordered field, and A may have any set-sized ordinal order type.

The hypothesis $A \subseteq G_{>0}$ is not a convenience. If 0 were allowed into A , then part (b) would fail immediately: appending any number of zeros to a word representing g produces another word

representing g , so every element of $M(A)$ would have infinitely many representations, and the regrouping arguments that use (b) would collapse. Part (a) would survive, but it is (b) that the substitution results need.

A.4 Finite convolution and formal substitution

Corollary A.4. *If $A, B \subseteq G$ are well-ordered, then $A + B$ is well-ordered, and for each $c \in G$ only finitely many pairs $(a, b) \in A \times B$ satisfy $a + b = c$.*

Proof. There is nothing to prove if either set is empty. Otherwise let a_0, b_0 be their minima. The set

$$C = ((A - a_0) \cup (B - b_0)) \setminus \{0\}$$

is well-ordered and consists of positive elements. A finite union of well-ordered subsets of a linear order is well-ordered, since the minimum of any nonempty subset is the smaller of its existing minima in the two parts.

Now $A + B \subseteq a_0 + b_0 + M(C)$, proving the first claim by Lemma A.3. For the second, translate a pair to its nonnegative components and remove any zero entries. It becomes a word of length at most two over C with sum $c - a_0 - b_0$. There are finitely many such words, and each word has only finitely many ways to restore the two labeled positions and possible zeros. Thus there are finitely many pairs. $\square$

The two-series case deserves a second, independent proof, because it is the only support fact used by the definition of the Hahn product itself. The following argument needs neither Higman's lemma nor the positive-support monoid, only Lemma A.1. Keeping it makes the *definition* of $\mathcal{H}_\uparrow(G)$ independent of any well-quasi-order machinery; the well-quasi-order machinery is needed only for infinite substitutions.

Lemma A.5 (Two supports, elementarily). *Let A, B be well-ordered subsets of a linearly ordered abelian group G . Then $A + B$ is well ordered, and every $c \in G$ has only finitely many representations $c = a + b$ with $(a, b) \in A \times B$.*

Proof. Suppose $c_0 > c_1 > c_2 > \dots$ were an infinite strictly decreasing sequence in $A + B$, and write $c_n = a_n + b_n$ with $a_n \in A$ and $b_n \in B$. By the second part of Lemma A.1 applied to $A \times B$, pass to a subsequence along which both a_n and b_n are nondecreasing. Then c_n is nondecreasing along that subsequence, contradicting strict decrease. In a linear order, the absence of an infinite strictly decreasing sequence gives well-ordering in the present set-theoretic setting: from a nonempty subset with no least element one recursively chooses a descending sequence.

For the second claim, suppose some c had infinitely many distinct representing pairs (a_n, b_n) . Again extract a subsequence that is coordinatewise nondecreasing. Two distinct pairs (a_i, b_i) and (a_j, b_j) in it satisfy $a_i \leq a_j$ and $b_i \leq b_j$ with at least one inequality strict, so $a_i + b_i < a_j + b_j$; but both sums equal c . $\square$

Corollary A.6 (Substitution into a positive-support series). *If $\epsilon \in \mathcal{H}_\uparrow(G)$ has support contained in $G_{>0}$, then for any real sequence $(d_n)_{n \geq 0}$ the Hahn sum*

$$\sum_{n \geq 0} d_n \epsilon^n$$

is well-defined. Sending a formal power series $\sum d_n z^n$ to this sum is a unital homomorphism $\mathbb{R}[[z]] \rightarrow \mathcal{H}_\uparrow(G)$.

Proof. For $\epsilon = 0$ this is the constant-term map. Otherwise apply Lemma A.3 to $A = \text{supp}(\epsilon)$. Every exponent occurring in a power lies in $M(A)$, and each exponent can occur in only finitely many powers because it has finitely many word representations, across all word lengths. Thus the family is summable.

To multiply two resulting sums, expand products as finite words in A . At a specified exponent there are finitely many words and finitely many splits of each word into two parts. Consequently the product can be regrouped by the two lengths, giving the ordinary formal power-series product. Preservation of addition and the unit is immediate. $\square$

This corollary justifies the binomial expressions in Lemma 5.1 without an analytic convergence argument. It also explains why setting the formal variable equal to an infinitesimal is natural: positive source exponent supports correspond, after reflection, to negative surreal normal-form exponents.

A.5 The field property

Proposition A.7. *For any linearly ordered abelian group G , $\mathcal{H}_1(G)$ is a field.*

Proof. By Corollary A.4, addition and multiplication are well-defined. Their ring identities follow from finite regrouping of coefficients; for three or more factors, iterate that corollary to see that each coefficient still has only finitely many contributing tuples.

Let $s \neq 0$ have leading term rx^a , where $r \neq 0$. Write $s = rx^a(1 + \epsilon)$ with $\text{supp}(\epsilon) \subseteq G_{>0}$. By Corollary A.6, the formal geometric identity gives

$$(1 + \epsilon) \sum_{n \geq 0} (-\epsilon)^n = 1.$$

Thus

$$s^{-1} = r^{-1}x^{-a} \sum_{n \geq 0} (-\epsilon)^n$$

is a well-defined inverse. All nonzero elements are invertible. $\square$

B Reproducing the artifact checks

The archive is self-contained apart from a standard LaTeX installation and Python 3.9 or later. It has no network-dependent verification steps and no third-party Python package requirements.

From the extracted archive directory, run

```
python code/verify.py --degree 256 --trials 1000 --output results
```

A successful run prints a JSON report with "status": "PASS" and rewrites the exact coefficient table and verification report. The explicit runtime checks are written with a require helper that raises a dedicated `VerificationFailure` exception, so they remain active even when Python is invoked with `-O`, and a failure exits with a nonzero status after printing a FAIL line rather than a traceback. The optional flags are `-degree`, `-trials`, `-fixed-pairs`, `-seed` and `-output`; the first three reject nonpositive values.

The file `results/coefficients.csv` has one header row and one row per degree $0 \leq n \leq 256$, with the seven columns `n`, `numerator`, `denominator` and `exact_fraction` for b_n , then `h_n`, `coefficient_B_squared` and `coefficient_H_squared`. The last two columns are the computed coefficients of B^2 and H^2 and should read $1, -1, 0, 0, \dots$ and $1, 1, 1, \dots$ respectively.

Artifact	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Standalone source and compiled article.
<code>short-proof.tex</code> , <code>short-proof.pdf</code>	One-page extraction of the decisive argument.
<code>code/verify.py</code>	Exact coefficient and finite-exponent checks.
<code>results/verification.json</code>	Machine-readable verification results and parameters.
<code>results/coefficients.csv</code>	Exact coefficients of both explicit witness series.
<code>references.bib</code>	Reusable bibliographic metadata; not a build dependency.
<code>proof-audit.md</code> , <code>sources.md</code>	Logical audit, and version-specific source record.
<code>README.md</code> , <code>build.sh</code>	Reproduction instructions and a local build command.

Build the article using

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

or the supplied `build.sh`. The document uses standard TeX Live packages and the bundled installation's Palatino-style newpx fonts; no font files are redistributed. The typeset bibliography is included directly in the LaTeX source, so no BibTeX run is required. Reusable bibliographic metadata are also supplied in `references.bib`.

The supplied `proof-audit.md` records the logical assumptions and the limits of the computational evidence. `sources.md` records inspected primary sources and the status-search boundary.

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